Quiz 5

1. Convolution

An N-point DFT multiplication of two sequences implements what kind of convolution by default?

N-point circular convolution.

2. Aliasing

A length-50 sequence is convolved with a length-17 FIR by FFT. What minimum transform length avoids time aliasing?

$50 + 17 - 1 = 66$ points minimum.

3. OLA vs OLS

In one sentence, distinguish overlap-add from overlap-save.

Non-overlapping inputs and overlapping outputs for overlap-add, overlapping inputs and outputs with first $L_h - 1$ output samples discarded for overlap-save.

4. STFT duration

An STFT uses a 256-sample window at $F_s = 8$ kHz. What is the window duration in milliseconds?

$t = 256/8 = 32$ ms

5. Frequency localization

Which generally improves frequency localization: a longer or shorter STFT window? What does it cost?

A longer window generally improves frequency localization/resolution but worsens time localization and can smear fast changes.

6. Power spectrogram

What quantity is visualized by a power spectrogram?

The squared magnitude of the STFT, $|X(r, \omega)|^2$, usually displayed on a linear or dB scale.